Local approximation widths for elliptic equations with measurable coefficients

Kaiwen Jin
yui.ui.siki@gmail.com

11 October 2026

Abstract

We prove that the local energy approximation widths of scalar, symmetric, uniformly elliptic equations in dimensions two and three decay as $\exp(-cn^{1/(d-1)})$, with no regularity assumption on the coefficients beyond measurability and boundedness. The result applies to the normalized cutoff operator on nested cubes in AIM Problem 517. Its proof compares the derivative of a harmonic energy Gram matrix with tangential boundary energy, and the Gram matrix itself with the energy of an ordinary harmonic extension. Spherical harmonic counting then bounds the determinant and the restriction singular values. A finite ball covering transfers the estimate to the prescribed cutoff. Exact propagation through radial layers supplies reproducible numerical checks of the determinant estimate. The constants may depend on coefficient contrast.

1 Introduction and main result

Local spaces of solutions of an elliptic equation provide coefficient-adapted basis functions for multiscale approximation. Their Kolmogorov widths measure the best error attainable with a prescribed local dimension, after allowing solutions on a larger domain. The cutoff formulation used in generalized finite element methods retains the partition of unity in the approximation norm; see Ma, Scheichl and Dodwell [7, Section 3.1]. For the scalar elliptic case, Ma's abstract theory gives decay with exponent $1/d$ [6, Theorem 3.8, Remark 3.9 and Section 6.1]; Section 7 of that preprint asks for the exponent $1/(d-1)$.

For ordinary harmonic functions, the boundary dimension already determines singular-value decay. In their author-hosted final text, Bandtlow and Chu give exact ball formulas in the L^2 harmonic Bergman setting and extend the decay to relatively compact domain pairs by ball covers [3, Proposition 3.3 and Theorem 4.5]. Measurable elliptic coefficients do not preserve that explicit harmonic expansion. We instead compare two quadratic forms on almost every sphere; the reference spherical harmonics enter only in a variational estimate. This yields the same exponent for the energy restriction and for the scalar cutoff widths asked for in AIM Problem 517 [1].

Let $d \in \{2,3\}$ and let $\omega \Subset \omega^* \subset \mathbb{R}^d$ be concentric open cubes. Suppose $A \in L^\infty(\omega^*;\mathbb{R}^{d\times d})$ is real symmetric and

$$\alpha|\xi|^2 \le \xi^T A(x)\xi \le \beta|\xi|^2, \qquad 0<\alpha\le\beta<\infty, \tag{1}$$

for almost every x and every $\xi \in \mathbb{R}^d$. For a subdomain D set

$$a_D(u,v)=\int_D A\nabla u\cdot\nabla v\,\mathrm{d}x, \qquad \|u\|_{a,D}=a_D(u,u)^{1/2}.$$

This is a seminorm until constants have been removed. Fix a nonzero $\chi \in W^{1,\infty}(\omega) \cap H^1_0(\omega)$ and define

$$\mathcal{H}_A(\omega^*) = \{u \in H^1(\omega^*) : a_{\omega^*}(u, v) = 0 \text{ for every } v \in H^1_0(\omega^*)\},$$
$$\mathcal{H}^0_A(\omega^*) = \{u \in \mathcal{H}_A(\omega^*) : a_\omega(\chi u, \chi) = 0\}.$$

For $Tu = \chi(u|_\omega)$, the width in the target energy norm is

$$d_n(T) = \inf_{\substack{V \subset H^1_0(\omega) \text{ linear} \\ \dim V \le n}} \sup_{\substack{u \in \mathcal{H}^0_A(\omega^*) \\ \|u\|_{a,\omega^*} \le 1}} \inf_{v \in V} \|Tu - v\|_{a,\omega}. \tag{2}$$

Theorem 1.1. *For every fixed choice of the preceding data, there are $C, c > 0$, independent of n, such that*

$$d_n(T) \le C \exp\bigl(-cn^{1/(d-1)}\bigr) \qquad \text{for every integer } n \ge 1.$$

The spaces, normalization and norms in (2) are identical to those in Problem 517, so Theorem 1.1 proves its proposed bound for every admissible coefficient and cutoff. No smoothness or scale separation of A is imposed. We do not claim contrast-independent constants or a matching lower bound for every cutoff. The exact ordinary harmonic ball model has this exponent, as described in Section 4.

The proof estimates the growth of the determinant of an energy Gram matrix on expanding balls. Boundary spectral estimates and energy determinants also appear in Li and Wang's dimension bounds for harmonic functions [5, Lemma 1 and Section 2]; we use a local comparison with the ordinary harmonic extension. The key inequalities are an upper bound for the interior energy by that extension energy, and a lower bound for its radial derivative by tangential boundary energy. Their combination avoids differentiating the coefficient. Almost-everywhere Sobolev slicing and a separate singular-Gram-matrix case make the argument applicable at the stated regularity. Section 2 proves the ball estimate, Section 3 transfers it to the cutoff, and Section 4 describes radial computations. All spaces are real, and surface integrals use ordinary area measure.

2 Energy restriction on balls

Write $B_t = B_t(0)$; translation gives balls with any center. Throughout this section A satisfies (1) on B_R. Let $X_t = \mathcal{H}_A(B_t)/\mathbb{R}$, equipped with its energy inner product. A class has a unique representative of zero ordinary mean on B_t. For $0 < r < R$ let

$$S_{r,R} : X_R \longrightarrow X_r, \qquad [u] \longmapsto [u|_{B_r}].$$

We will bound its singular values, listed in nonincreasing order and completed with zeros when the rank is finite.

We use the following standard Sobolev facts on bounded Lipschitz domains: H^1 is complete; smooth functions are dense; there is a bounded extension to $H^1(\mathbb{R}^d)$; $H^1(D) \to L^2(D)$ is compact in the dimensions considered here; the trace $H^1(D) \to H^{1/2}(\partial D)$ is continuous with kernel $H^1_0(D)$; and

$$\|v - v_D\|_{L^2(D)} \le C_D \|\nabla v\|_{L^2(D)}, \qquad v_D = \frac{1}{|D|} \int_D v. \tag{3}$$

On $H^1_0(D)$ the estimate holds without subtracting v_D. Precise statements are in Guermond's notes [4, Proposition 2.9, Theorems 2.21, 2.30, 2.35 and 3.10, Lemmas 3.24 and 3.27]. Balls and

cubes satisfy the domain assumptions; on B_t, scaling gives $C_D = C_P t$, where C_P depends only on dimension. Multiplication by a Lipschitz function is continuous on H^1, with weak derivative $\nabla(\eta u) = \eta \nabla u + u \nabla \eta$. Prove this first for smooth u and pass by density and the product bound. Trace continuity gives the boundary product identity.

Lemma 2.1. *Each X_t is a separable Hilbert space and $S_{r,R}$ is compact and contractive.*

Proof. On zero-mean representatives, (3) and ellipticity make the H^1 and energy norms equivalent. The weak harmonic equations and zero mean define a closed subspace of $H^1(B_t)$, proving completeness. Restriction cannot increase the energy. For $r < s < R$ take a Lipschitz cutoff η supported in B_s, equal to one on B_r, with $|\nabla \eta| \le (s-r)^{-1}$. Testing the weak equation for w with $\eta^2 w$, and using Cauchy–Schwarz, gives

$$\int \eta^2 A \nabla w \cdot \nabla w \le 2 \left(\int \eta^2 A \nabla w \cdot \nabla w \right)^{1/2} \left(\int w^2 A \nabla \eta \cdot \nabla \eta \right)^{1/2}.$$

Division when the first factor is nonzero, and the zero case, give

$$\|w\|_{a,B_r} \le \frac{2\sqrt{\beta}}{s-r} \|w\|_{L^2(B_s)}. \tag{4}$$

A bounded sequence in X_R, represented with zero B_R mean, is bounded in $H^1(B_R)$. Rellich gives an $L^2(B_R)$-convergent subsequence. Apply (4) to its pairwise differences: the restrictions are Cauchy in X_r, hence converge there. □

2.1 Boundary forms for the ordinary harmonic extension

Let Y_j be a real $L^2(S^{d-1})$ orthonormal basis of nonconstant spherical harmonics, ordered by their degrees $\ell_j \ge 1$. The degree ℓ has multiplicity 2 in $d = 2$ and $2\ell + 1$ in $d = 3$. These dimension and completeness facts follow from [2, Proposition 5.8 and Theorem 5.12]; its probability surface measure is converted to area measure by rescaling the basis. Applying the polar Laplacian to $\rho^\ell Y$ gives $-\Delta_S Y = \ell(\ell + d - 2) Y$.

For boundary data $f(\theta)$ on ∂B_t, modulo constants, define its ordinary harmonic extension h_f and the two forms

$$b_t(f) = \int_{B_t} |\nabla h_f|^2, \qquad q_t(f) = \int_{\partial B_t} |\nabla_\tau f|^2. \tag{5}$$

The second form is defined for $f \in H^1(S^{d-1})/\mathbb{R}$; its tangential gradient is on the sphere of radius t. If $f = \sum_j c_j Y_j$ after subtracting its surface mean, then

$$b_t(f) = t^{d-2} \sum_j \ell_j c_j^2, \qquad q_t(f) = t^{d-3} \sum_j \ell_j (\ell_j + d - 2) c_j^2. \tag{6}$$

We give the regularity justification since these forms will be used on Sobolev slices. For smooth f, integration by parts on the sphere and L^2 completeness give

$$\int_S |\nabla_S f|^2 = \langle f, -\Delta_S f \rangle = \sum_j \ell_j (\ell_j + d - 2) c_j^2.$$

The series pairing is justified by Cauchy–Schwarz between the coefficients of the two L^2 functions. Smooth density on the sphere follows by local charts and mollification. Applying the identity to

differences of smooth approximations proves it for H^1 data and proves convergence of spherical partial sums in $H^1(S)$. The extension series

$$h_f(\rho\theta) = \sum_j c_j (\rho/t)^{\ell_j} Y_j(\theta)$$

converges in $H^1(B_t)$. For finite sums its energy is the first formula in (6), and its squared L^2 norm is $t^d \sum_j c_j^2/(2\ell_j + d)$. Both series converge by the preceding identity. Each mode has energy $\ell_j t^{d-2}$ by radial integration; angular orthogonality removes cross terms. Trace continuity identifies the limit's trace with f, and H^1 convergence preserves weak harmonicity. A constant boundary component extends as a constant. This proves (6).

Put

$$\nu_j = \ell_j + d - 2, \qquad \Sigma_d(m) = \sum_{j=1}^{m} \nu_j. \tag{7}$$

The positive generalized eigenvalues of (q_t, b_t) are ν_j/t. The multiplicities give

$$\Sigma_2(m) \geq \frac{m^2}{4}, \qquad \Sigma_3(m) \geq \frac{2}{3} m^{3/2}. \tag{8}$$

For $d = 2$, $\ell_j = \lceil j/2 \rceil \geq j/2$. For $d = 3$, there are $(\ell+1)^2 - 1$ positive modes through degree ℓ, so $\nu_j \geq \sqrt{j}$ and the sum dominates $\int_0^m \sqrt{x}\, \mathrm{d}x$.

Lemma 2.2. *If $f_1, \ldots, f_m \in H^1(S^{d-1})/\mathbb{R}$ are independent, and B, Q are their Gram matrices for b_t, q_t, then*

$$\operatorname{tr}(B^{-1}Q) \geq \frac{\Sigma_d(m)}{t}.$$

Proof. Change basis to b_t-orthonormal functions $g_1, \ldots, g_m$. In the b_t-orthonormal basis $e_j = (t^{d-2}\ell_j)^{-1/2} Y_j$ write $g_i = \sum_j z_{ji} e_j$. The weights $w_j = \sum_i z_{ji}^2$ satisfy $0 \leq w_j \leq 1$ by Bessel's inequality and $\sum_j w_j = m$ by orthonormality. Thus

$$\operatorname{tr}(B^{-1}Q) = \sum_i q_t(g_i) = \frac{1}{t} \sum_j \nu_j w_j \geq \frac{1}{t} \sum_{j=1}^{m} \nu_j.$$

The missing mass $\sum_{j \leq m} (1 - w_j)$ equals $\sum_{j>m} w_j$. Every tail weight has eigenvalue at least ν_m, whereas each missing weight in the first m has eigenvalue at most ν_m. The series are nonnegative, so this comparison holds also for their infinite limits. □

2.2 The energy determinant

Lemma 2.3. *Let $u_1, \ldots, u_m \in \mathcal{H}_A(B_R)$ and $G(t) = (a_{B_t}(u_i, u_j))_{i,j=1}^m$. If $G(R)$ is positive definite, then for $0 < r < R$,*

$$\frac{\det G(r)}{\det G(R)} \leq \left(\frac{r}{R}\right)^{(\alpha/\beta)\Sigma_d(m)}. \tag{9}$$

Proof. Each energy density is integrable. Polar integration and Fubini give absolute continuity of G on $[0, R]$ and

$$G'_{ij}(t) = \int_{\partial B_t} A \nabla u_i \cdot \nabla u_j \quad \text{for almost every } t. \tag{10}$$

The Lebesgue integration statements are [4, Theorems 1.47 and 2.2].

At almost every radius the traces of the finite family belong to $H^1(S^{d-1})$, with tangential gradients given by the tangential projections of their weak gradients. To justify this, on an annulus bounded away from zero use finite smooth sphere charts. The weak coordinate rule [4, Corollary 2.25] and Fubini give the angular Sobolev slices. More specifically, a smooth H^1 approximation converges in $L^2((a,b);H^1(S^{d-1}))$ for $0<a<b<R$; a subsequence converges in $H^1(S)$ at almost every radius. For every fixed $t<b$, the same approximation converges in $H^1(B_t)$, hence its traces converge in $L^2(\partial B_t)$. The two limits agree at good radii. A countable cover by annuli and the finite intersection for the u_i give one full-measure set in (r,R).

At a good radius let $B(t), Q(t)$ be the boundary-form Gram matrices of these traces modulo constants. Ellipticity gives

$$G'(t) \geq \alpha Q(t), \qquad G(t) \leq \beta B(t). \tag{11}$$

The first inequality follows by discarding the normal component from $\alpha \int_{\partial B_t} |\nabla \sum_i z_i u_i|^2$. For the second, let $u = \sum_i z_i u_i$ and let h be the ordinary harmonic extension of its trace. The preceding construction gives $h \in H^1(B_t)$ with the same trace, so $h-u \in H^1_0(B_t)$. Weak harmonicity on B_t follows by extending its tests by zero into B_R and implies

$$a_{B_t}(h,h) = a_{B_t}(u,u) + a_{B_t}(h-u, h-u) \geq a_{B_t}(u,u).$$

Use $a_{B_t}(h,h) \leq \beta \int_{B_t} |\nabla h|^2$ to conclude. No derivative or classical boundary trace of A is used.

If $G(r)$ is singular, (9) holds with zero left side. Suppose $G(r) > 0$. Monotonicity gives $G(t) \geq G(r) > 0$ on $[r,R]$. At a good radius the boundary traces are independent modulo constants: a constant boundary combination u would give $u - c \in H^1_0(B_t)$, and testing with $u-c$ would give zero energy, contradicting positive definiteness unless the combination is zero. Hence $B(t) > 0$. Inverting (11), using positivity under the trace, and applying Lemma 2.2, gives

$$\operatorname{tr}(G^{-1}G') \geq \alpha \operatorname{tr}(G^{-1}Q) \geq \frac{\alpha}{\beta} \operatorname{tr}(B^{-1}Q) \geq \frac{\alpha}{\beta} \frac{\Sigma_d(m)}{t}. \tag{12}$$

The middle step uses $G^{-1} - \beta^{-1}B^{-1} \geq 0$ and the nonnegative trace of its product with $Q \geq 0$.

The entries of G are absolutely continuous and its eigenvalues are bounded below on $[r,R]$. On the convex hull of its range, $\log\det$ has bounded derivative, hence is Lipschitz. Its composition with G is therefore absolutely continuous. At differentiability points the cofactor formula gives $(\log\det G)' = \operatorname{tr}(G^{-1}G')$. Integrating (12) proves (9). $G(R)$ is the continuous endpoint of the volume integral, so R need not be a good slice. The argument uses Dirichlet uniqueness on the same ball, and does not assume unique continuation between balls. □

Theorem 2.4. *The singular values s_m of $S_{r,R}$ satisfy*

$$\prod_{j=1}^{m} s_j^2 \leq (r/R)^{(\alpha/\beta)\Sigma_d(m)}, \qquad s_m \leq \exp\left[-\frac{\alpha}{2\beta}\log(R/r)\frac{\Sigma_d(m)}{m}\right] \quad (m \geq 1). \tag{13}$$

Consequently $d_k(S_{r,R}) = s_{k+1} \leq \exp[-b(k+1)^{1/(d-1)}]$ for $k \geq 0$, where

$$b = \frac{\alpha}{8\beta}\log(R/r) \quad (d=2), \qquad b = \frac{\alpha}{3\beta}\log(R/r) \quad (d=3). \tag{14}$$

Proof. Lemma 2.1 permits a singular decomposition. Apply the compact self-adjoint spectral theorem [8, p.1] to the nonnegative compact operator $S^*_{r,R}S_{r,R}$; Riesz representation [4, Theorem A.16] supplies the adjoint. The complex-space statements have real versions by complexification and a real orthonormal basis in each conjugation-invariant eigenspace. If $s_m > 0$, choose the first m

right singular vectors. Their outer energy Gram matrix is I and their inner one is $\operatorname{diag}(s_1^2, \dots, s_m^2)$, so Lemma 2.3 proves the product inequality. If $s_m = 0$, the product is zero. Since $s_m^{2m} \le \prod_{j=1}^m s_j^2$, the individual estimate follows; (8) then gives (14).

The first k left singular vectors approximate every image with error at most s_{k+1} times its input norm. Conversely, if $s_{k+1} > 0$ and $\dim V \le k$, there is a unit vector in the span of the first $k+1$ right vectors whose image is orthogonal to V. Its image norm is at least s_{k+1}. If $s_{k+1} = 0$, at most k left directions already span the range. This proves the width identity, including finite rank and $k = 0$. □

3 Cutoff widths on nested cubes

Set $q = A\nabla\chi \cdot \nabla\chi$ on ω. Then $q \ge 0$, $q \in L^\infty(\omega)$, and $Q_\chi = \int_\omega q = a_\omega(\chi, \chi) > 0$ by ellipticity and zero-trace Poincare. For harmonic u, v, the cutoff identity is

$$a_\omega(\chi u, \chi v) = \int_\omega quv. \tag{15}$$

It also appears in [7, Lemma 3.1]. Test the two weak equations with the zero extensions of $\chi^2 v$ and $\chi^2 u$, admissible by the product and trace rules. Adding the equations and using symmetry cancels precisely the gradient and cross terms in the expansion of the left side of (15). Taking $v = 1$ shows that $\mathcal{H}_A^0$ is exactly $\int_\omega qu = 0$. If $\bar{u}$ is the ordinary mean on ω^*, then

$$Q_\chi |\bar{u}| \le \|q\|_{L^2(\omega)} \|u - \bar{u}\|_{L^2(\omega^*)}.$$

Mean Poincare and ellipticity yield

$$\|u\|_{H^1(\omega^*)} \le C_{\mathrm{norm}} \|u\|_{a,\omega^*} \quad (u \in \mathcal{H}_A^0(\omega^*)). \tag{16}$$

The weak equations and weighted normalization are closed constraints in H^1. Thus $\mathcal{H}_A^0$ is Hilbert in the energy norm, and (15) and (16) make T bounded. Rellich and (15) on differences also make T compact.

Proof of Theorem 1.1. Let $\delta = \operatorname{dist}(\overline{\omega}, \partial\omega^*) > 0$ and $r = \delta/8$. Choose centers $x_1, \dots, x_M \in \overline{\omega}$ whose $B_{r/2}(x_j)$ cover $\overline{\omega}$. Each $B_{2r}(x_j)$ lies in ω^*. Fix smooth functions ψ_j supported in $B_r(x_j)$ with sum one near $\overline{\omega}$. Such a partition is obtained from nonnegative bumps ρ_j equal to one on the half-balls: let $F = \sum_j \rho_j$, choose $\zeta \in C_c^\infty(\{F > 0\})$ equal to one near $\overline{\omega}$, and extend $\psi_j = \zeta\rho_j/F$ by zero. The support of ζ stays away from the zero denominator.

The zero extension of χ is globally Lipschitz. Indeed, its Lipschitz representative on the closed convex cube has zero boundary values, and a segment joining an inside to an outside point meets that boundary first. Set $\phi_j = \chi\psi_j$. For a zero-mean $e \in H^1(B_r(x_j))$, the product rule and scaled Poincare give

$$\|\phi_j e\|_{a,\omega} \le D_j \|e\|_{a,B_r(x_j)}, \qquad D_j = \sqrt{\beta/\alpha}\big(\|\phi_j\|_\infty + C_P r \|\nabla\phi_j\|_\infty\big). \tag{17}$$

The product, extended by zero from the ball, lies in $H_0^1(\omega)$. There is no ball-boundary jump because of compact support, and its cube trace vanishes by the trace product rule.

For each ball let $W_{j,k}$ be the first k left singular directions of $S_{r,2r}$, represented with zero $B_r(x_j)$ mean. If the rank is smaller, use the whole range. Let $\Pi_{j,k}$ be the corresponding energy projection

on the quotient. Define $a_j(u)$ by adding the ordinary mean of u on $B_r(x_j)$ to the zero-mean representative of $\Pi_{j,k}[u|_{B_r(x_j)}]$. Then $e_j = u - a_j(u)$ has zero mean, and Theorem 2.4 gives

$$\|e_j\|_{a,B_r(x_j)} \le e^{-b(k+1)^p}\|u\|_{a,B_{2r}(x_j)} \le e^{-b(k+1)^p}\|u\|_{a,\omega^*}, \qquad p = \frac{1}{d-1},$$

with b fixed by (14) and $R/r = 2$. The space

$$V_k = \sum_{j=1}^{M} \{\phi_j(c+w) : c \in \mathbb{R},\ w \in W_{j,k}\} \subset H_0^1(\omega)$$

is independent of u and has dimension at most $M(k+1)$. With $v_u = \sum_j \phi_j a_j(u) \in V_k$, we have

$$Tu - v_u = \sum_j \phi_j e_j, \qquad \|Tu - v_u\|_{a,\omega} \le C_0 e^{-b(k+1)^p}\|u\|_{a,\omega^*}, \qquad C_0 = \sum_j D_j.$$

We used the product estimate for the errors, which need not be harmonic. For $n \ge M$ choose $k+1 = \lfloor n/M \rfloor \ge n/(2M)$. Then $\dim V_k \le n$ and the bound holds with $c = b/(2M)^p$, $C = C_0$. For $1 \le n < M$, boundedness of T permits increasing C to $\max\{C_0, \|T\|e^{cM^p}\}$. All covering functions and constants were fixed before n was chosen. □

4 Radial models and numerical checks

These computations concern ball energy restrictions. They check the determinant comparison on a finite family and are not premises of Theorem 1.1. For $A = I$, the mode $\rho^\ell Y(\theta)$, with Y normalized in area measure, has energy $\ell t^{2\ell+d-2}$ on B_t. Its restriction singular value is therefore

$$\sigma_\ell = (r/R)^{\ell+(d-2)/2}. \tag{18}$$

These are all the nonconstant modes: angular slicing of a weak harmonic function makes each coefficient solve the radial Laplace ODE. For $\ell \ge 1$ its solutions are ρ^ℓ and $\rho^{-(\ell+d-2)}$; the second has infinite energy at zero. In degree zero only a constant survives, since ρ^{2-d}, or $\log\rho$ in two dimensions, has infinite energy. Angular and radial Parseval identities, integrated on annuli and passed by nonnegative limits to the ball, give completeness in the quotient energy norm. Multiplicity counting then gives $-\log s_m \asymp m^{1/(d-1)}$. The power of r/R here is $\ell + (d-2)/2$, whereas the L^2 Bergman power is $\ell + d/2$ in [3, Proposition 3.3]. Both give the same dimension-dependent power $1/(d-1)$ in the stretched-exponential width decay.

4.1 Exact layer propagation

Let

$$A(\rho\theta) = a_r(\rho)\theta\theta^T + a_t(\rho)(I - \theta\theta^T),$$

where a_r, a_t are positive and piecewise constant on finitely many concentric layers, including a constant core $0 < \rho < \rho_0$. The value at zero is immaterial. For degree $\ell \ge 1$ put $\lambda = \ell(\ell+d-2)$. Separation of variables in the weak equation gives

$$(\rho^{d-1}a_r f')' - a_t\lambda\rho^{d-3}f = 0, \qquad f \text{ and } \rho^{d-1}a_r f' \text{ continuous at interfaces.} \tag{19}$$

For $t = \log\rho$ and $q = a_r\rho f'$, the state on one layer satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} f \\ q \end{pmatrix} = M\begin{pmatrix} f \\ q \end{pmatrix}, \qquad M = \begin{pmatrix} 0 & 1/a_r \\ a_t\lambda & -(d-2) \end{pmatrix}. \tag{20}$$

Both entries are continuous at interfaces. Finite core energy selects $f \propto \rho^{\gamma_+}$, where

$$\gamma_+ = \frac{\sqrt{(d-2)^2 + 4(a_t/a_r)\lambda} - (d-2)}{2} > 0, \qquad \gamma_- = -(d-2) - \gamma_+.$$

Normalize $(f, q) = (1, a_r\gamma_+)$ at ρ_0. Multiplication of (19) by f and integration through the layers gives

$$E_\ell(\rho) = \int_0^\rho \left[a_r (f')^2 + \frac{a_t \lambda f^2}{s^2} \right] s^{d-1}\, \mathrm{d}s = \rho^{d-2} f(\rho) q(\rho). \tag{21}$$

Interface terms cancel, and the term at zero vanishes because $2\gamma_+ + d - 2 > 0$. Hence $\log \sigma_\ell = \frac{1}{2}(\log E_\ell(r) - \log E_\ell(R))$.

Direct unscaled transfer matrices can overflow. Use the exact formula

$$e^{M\Delta t} = e^{\gamma_+ \Delta t}[e^{-g\Delta t} I + (1 - e^{-g\Delta t}) P_+], \qquad g = \gamma_+ - \gamma_-, \quad P_+ = \frac{M - \gamma_- I}{g}. \tag{22}$$

The bracket has nonnegative entries. Remove the scalar exponential and normalize the positive state after each segment, accumulating its logarithmic scale. Computing $1 - e^{-g\Delta t}$ with $-\operatorname{expm1}(-g\Delta t)$ avoids cancellation for short segments. The rationalized formula

$$\gamma_+ = \frac{2(a_t/a_r)\lambda}{\sqrt{(d-2)^2 + 4(a_t/a_r)\lambda} + d - 2}$$

avoids subtraction in the exponent.

Algorithm 1 Logarithmic energy ratios through radial layers

Require: $d \in \{2, 3\}$; positive layer coefficients with outer edges $0 < \rho_0 < \cdots < \rho_J = R$; $0 < r < R$; maximal degree $L \geq 1$

Ensure: $\log \sigma_\ell$ for $\ell = 1, \ldots, L$

for $\ell = 1, \ldots, L$ **do**
  Compute the core γ_+; set $z = (1, a_r\gamma_+)^T$
  $h \leftarrow \log \|z\|_\infty$; $z \leftarrow z/\|z\|_\infty$
  $F(\rho_0) \leftarrow (d-2)\log \rho_0 + \log(a_r\gamma_+)$
  if $r \leq \rho_0$ **then**
    $F(r) \leftarrow F(\rho_0) + (2\gamma_+ + d - 2)\log(r/\rho_0)$
  end if
  for each remaining layer in order **do**
    Split at r if it lies strictly inside this layer
    for each resulting segment $[a, b]$ in order **do**
      Compute g, P_+, γ_+ for this layer; $\Delta t \leftarrow \log(b/a)$
      $z \leftarrow [e^{-g\Delta t} I + (1 - e^{-g\Delta t}) P_+] z$
      $h \leftarrow h + \gamma_+ \Delta t + \log \|z\|_\infty$; $z \leftarrow z/\|z\|_\infty$
      $F(b) \leftarrow (d-2)\log b + \log z_1 + \log z_2 + 2h$
    end for
  end for
  $\log \sigma_\ell \leftarrow (F(r) - F(R))/2$
end for

Algorithm 1 costs $O(L(J+1))$ operations and $O(J+L)$ storage for inputs and outputs, with $O(1)$ additional state storage. Expanding and sorting the included K modes costs $O(K \log K)$ operations

and $O(K)$ storage, where $K = 2L$ in $d = 2$ and $K = L(L+2)$ in $d = 3$. These counts exclude plotting and high-precision reference calculations. There is no spatial quadrature or time-step error for these layer data; floating-point roundoff remains. A finite-degree sort does not certify that omitted degrees cannot interlace.

4.2 Cases, metrics and validation

The campaign uses seed 5172026, $R = 1$, $r = 1/2$, a core ending at 2^{-8}, 64 logarithmically spaced annuli, and degrees 1 through 80. Each dimension has 20 cases: isotropic $a_r = a_t = 1$; constant polar anisotropy $a_r = 4, a_t = 1/4$; alternating $a_r \in \{1/5, 5\}$ with $a_t = 1/a_r$; and 17 cases with independent $\exp(U[-3, 3])$ coefficient draws on each layer. The local manifest records all input arrays and results. Figure 1 displays the first four cases.

For an included spectrum $\widehat{s}_1 \geq \cdots \geq \widehat{s}_K$, let

$$L_m = 2\sum_{j=1}^{m} \log \widehat{s}_j, \qquad B_m = (\alpha/\beta)\log(r/R)\Sigma_d(m).$$

The determinant bound requires $L_m \leq B_m < 0$. Figure 2 plots $L_m/B_m \geq 1$; a larger value means more decay than the upper bound requires. Lemma 2.3 applies to every finite orthonormal harmonic family, so this finite-mode comparison is valid without certifying the full infinite ordering.

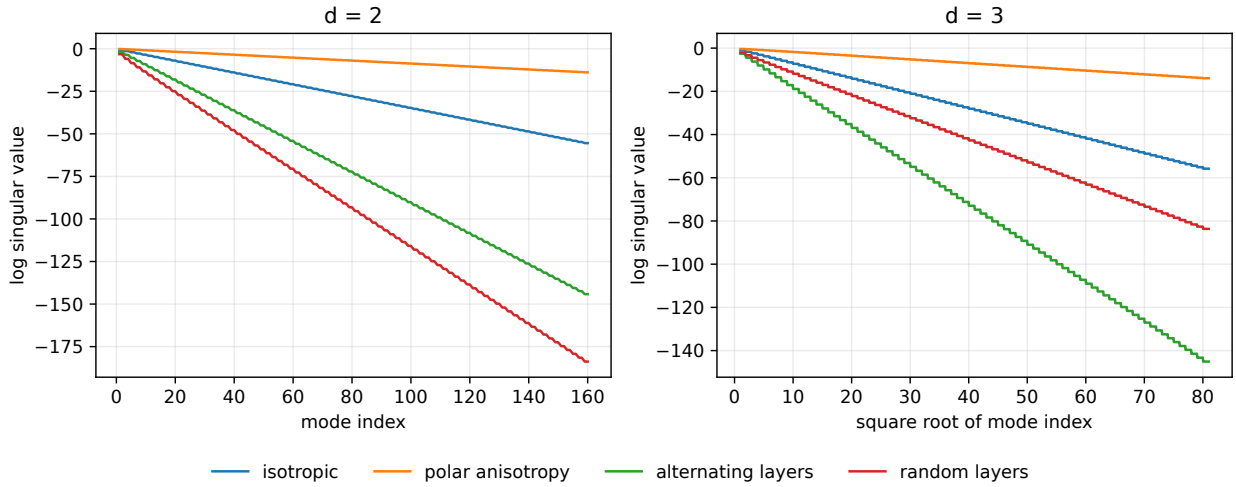


Figure 1: Natural logarithms of finite-degree energy singular values for four radial cases, against m or $\sqrt{m}$. Smaller values mean stronger contraction. All modes through degree 80 are shown with multiplicity; no tail ordering is inferred. No values are clipped or replaced by a floor. These are numerical checks on a finite radial family.

The implementation is compared with exact constant-coefficient formulas, independent unscaled 70-decimal-digit matrix exponentials, and subdivision of each annulus into two identical coefficient layers. Subdivision tests representation invariance, rather than mesh convergence. The independent reference constructs the matrix exponential directly and does not call (22). It covers both dimensions, varying contrasts and inner radii in the core, within annuli and on interfaces. The library has 23 unit tests, including coefficient and geometric scaling, invalid input rejection, spherical multiplicities and independently derived generalized-eigenvalue examples.

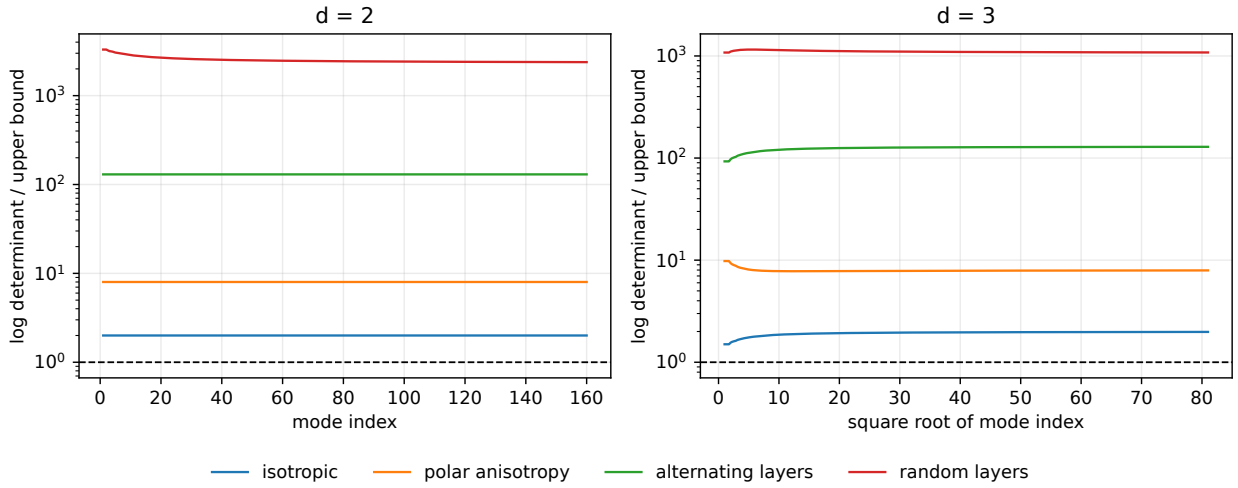


Figure 2: The negative-log ratio L_m/B_m for the same included modes. Values above the dashed line at one satisfy the determinant bound. The vertical axis is logarithmic and the horizontal axis is m or $\sqrt{m}$. Both numerator and denominator are negative; larger ratios indicate a stronger inequality. Every included prefix is plotted, without clipping or a floor.

There are also 40 finite-dimensional form comparisons per dimension. In a sphere-mode subspace of ambient dimension 40, random orthonormal columns form B, Q using (6). A positive matrix $G \le \beta B$ and $E' \ge \alpha Q$ are constructed with $\alpha = 0.15$, $\beta = 4$. Cholesky whitening computes $\operatorname{tr}(G^{-1}E')$ and checks (12). This tests the finite matrix inequality under its premises; it does not simulate a general rough-coefficient PDE.

Table 1: Recorded binary64 checks with Python 3.13.3 and NumPy 2.4.4. The independent reference uses mpmath at 70 decimal digits. Errors are absolute differences of logarithmic singular values.

Quantity	Observed value
Radial cases / finite-form trials	40 / 80
Largest exact constant-case error	1.71×10^{-13}
Largest layer-subdivision difference	1.02×10^{-12}
Largest determinant excess $\max(L_m - B_m)$	-0.30325
Smallest finite-form trace margin	1.46480

All 40 cases passed the determinant and individual bounds with logarithmic tolerance 10^{-8}; subdivision tolerance was 2×10^{-11}. Exact and high-precision tolerances are specified per test in the code. Table 1 records reproducible numerical evidence, not interval-certified errors or a proof for all coefficients. The prescribed layer distribution is not a statement about typicality in the full coefficient class.

The accompanying package contains the library, unit tests, case configuration, scripts, figures and compact summaries. Detailed arrays and an environment manifest are regenerated locally. From its root run `python3 -m unittest discover -s test -v` with `PYTHONPATH=src`, then `python3 numerical_case/run_cases.py` and `python3 numerical_case/plot_cases.py`. The theorem is independent of these computations.

References

[1] 517. sharp local approximation widths for rough elliptic equations. AIM problem catalogue, MColbrook/AIM, https://github.com/MColbrook/AIM/blob/c9929805cece044705abaaebe1cebddac1ba11a6/problems/517-sharp-local-elliptic-widths.md. Commit c9929805cece044705abaaebe1cebddac1ba11a6.

[2] Sheldon Axler, Paul Bourdon, and Wade Ramey. *Harmonic Function Theory*. Springer, second edition, 2001. Author-hosted electronic version dated 17 July 2020, https://www.axler.net/HFT.pdf.

[3] Oscar F. Bandtlow and Cho-Ho Chu. Eigenvalue decay of operators on harmonic function spaces. *Bulletin of the London Mathematical Society*, 41(5):903–915, 2009. https://doi.org/10.1112/blms/bdp068. Numbered locators refer to the author-hosted final text, https://webspace.maths.qmul.ac.uk/o.bandtlow/publ/harmonic_final.pdf.

[4] Jean-Luc Guermond. MATH 661. Lecture notes, Texas A&M University, https://people.tamu.edu/~guermond/M661_FALL_2025/index.html. Chapters 1–3 and Appendix A; online archive M661_FALL_2025, accessed 11 October 2026.

[5] Peter Li and Jiaping Wang. Counting dimensions of L-harmonic functions. *Annals of Mathematics*, 152(2):645–658, 2000. https://doi.org/10.2307/2661394.

[6] Chupeng Ma. A unified framework for multiscale spectral generalized FEMs and low-rank approximations to multiscale PDEs. https://arxiv.org/abs/2311.08761v3, 2024. arXiv:2311.08761v3, 17 December 2024.

[7] Chupeng Ma, Robert Scheichl, and Tim Dodwell. Novel design and analysis of generalized FE methods based on locally optimal spectral approximations. https://arxiv.org/abs/2103.09545v3, 2021. arXiv:2103.09545v3, 21 December 2021.

[8] Richard Melrose. 18.155 lecture 14. Massachusetts Institute of Technology, https://math.mit.edu/~rbm/18-155-F17/L14.pdf, 2017. 31 October 2017.

AI use statement

ChatGPT supplied an initial proof candidate and literature notes. Codex expanded the natural-language proof, checked cited source passages, prepared and tested the numerical implementation, and assisted with writing and typesetting. The proof and procedures are given in full for independent checking. These AI checks do not constitute independent human review or formal verification.